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Paper Code : PCC-CS503/PCCS503/PCCS503/PCCIB502 Object Oriented Programming

UPID : 005504

Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) The Inheritance property is used when there is an IS-A relationship between classes: TRUE or FALSE.
- (II) Which container class doesn't contain title bar and menu bars but it can have other components like button, text field etc.?
- (III) A class member declared protected becomes a member of subclass of which type?
- (IV) Which keyword is used to access the features of a package?
- (V) Which of the following statements are true for interfaces in java?
 - i) Objects of an interface cannot be created.
 - ii) All the methods of an interface are either abstract or default or static.
 - iii) It does not contain constructors.
- (VI) In the model-view-controller (MVC) architecture, the model defines the _____ layer.
- (VII) All collection classes are available in _____ package.
- (VIII) What would be the possible output if a constructor is declared with a return type?
 - a) Compilation error.
 - b) Runtime error.
 - c) Compilation and runs successfully
 - d) Only String return type is allowed
- (IX) main() method can be overloaded in Java. The above statement is true or false?
- (X) A class inheriting an abstract class does not define or implement all of its abstract methods is called _____ class.
- (XI) What types of polymorphism is used in the following java program.


```
class Paint {
    public void Color(int x) {...}
    public void Color(int x, int y) {...}
    public void Color(int x, int y, int z) {...}
}
```
- (XII) What is the output of following Java code?


```
class Test {
    private int a = 4;
    private int b = 1;
    void getSum(int a, int b) {
        a = a;
        b = b;
        System.out.println(this.a + this.b);
    }
}
public class Main {
    public static void main(String args[]) {
        Test T = new Test();
        T.getSum(3, 5);
    }
}
```

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. Explain different features of object oriented programming?

[5]

3. What are the differences between Procedural programming language and Object oriented programming language? [5]
4. What is the use of static keyword in java?
Write a program to explain the use of static variable and static method in java. [5]
5. Explain the benefit of using interface in java with the help of an example. [5]
6. Why multiple inheritance can be implemented by using Interfaces but not by classes? Explain with an example. [5]

Group-C (Long Answer Type Question)

Answer any three of the following

[15 x 3 = 45]

7. (a) What is parametric and non parametric constructor? Explain with suitable example. [5]
(b) Illustrate the use of the this keyword with a suitable program. [5]
(c) What is constructor overloading? Explain it with a suitable example. [5]
8. (a) What are the advantages of using design pattern in an organization? [3]
(b) Explain different types of design patterns with their use. [6]
(c) Explain the concept of model view controller pattern. What are the advantage of using it? [4+2]
9. (a) Create a class shape with data members dimension1, dimension2, area. Write a abstract method calculate_area in it. Create two child classes circle, triangle and override the calculate_area method in the child classes. [6]
(b) How run time polymorphism is achieved by using method overriding explain with respect to the problem statement given in part (a). [5]
(c) What is super keyword. Explain its use with an example program. [4]
10. (a) Write a program using swing to add two numbers and display the result. [7]
(b) Explain the working mechanism of Event delegation model. [5]
(c) Write some differences between Swing and AWT. [3]
11. (a) Explain the concept of Iterator design pattern. [4]
(b) Explain its working with a suitable program. [2+5]
(c) What is a final method? explain with an example. [4]

*** END OF PAPER ***